

0478/11 – JUNE 2019

Question	Answer	Marks
7	1 mark for each correct term, in the correct place: • Syntax • High-level language • Translator • Machine code • Assembly • Low-level language	6

0478/13 – JUNE 2019

Question	Answer	Marks
2(c)(i)	– Interrupt	1
2(c)(ii)	Two from: – Provides an interface – Loads / opens / installs / closes software – Manages the hardware // manages peripherals // spooling – Manages the transfer of programs into and out of memory – Divides processing time // processor management – Manages file handling – Manages error handling // interrupt handling – Manages security software – Manages utility software – Manages user accounts – Multitasking – Multiprogramming // time slicing – Batch processing	2

0478/11 – NOVEMBER 2019

Question	Answer	Marks
6(a)(i)	One from: ∞ Code will run without the need of an interpreter ∞ (Object) Code is platform independent ∞ Source code not available / cannot be modified	1
6(a)(ii)	One from: ∞ Source code not available / cannot be modified ∞ Comments, etc. not visible ∞ Future changes will require code to be recompiled	1

0478/13 – NOVEMBER 2019

Question	Answer	Marks															
2(a)	One mark for each correct row <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 60%;">Statement</th><th style="width: 10%;">True (✓)</th><th style="width: 10%;">False (✓)</th></tr> </thead> <tbody> <tr> <td>High-level languages need to be translated into machine code to run on a computer</td><td style="text-align: center;">✓</td><td></td></tr> <tr> <td>High-level languages are written using mnemonic codes</td><td></td><td style="text-align: center;">✓</td></tr> <tr> <td>High-level languages are specific to the computer's hardware</td><td></td><td style="text-align: center;">✓</td></tr> <tr> <td>High-level languages are portable languages</td><td style="text-align: center;">✓</td><td></td></tr> </tbody> </table>	Statement	True (✓)	False (✓)	High-level languages need to be translated into machine code to run on a computer	✓		High-level languages are written using mnemonic codes		✓	High-level languages are specific to the computer's hardware		✓	High-level languages are portable languages	✓		4
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0478/12 – MARCH 2020

Question	Answer				Mark
4					5
	Statement	Assembler (✓)	Compiler (✓)	Interpreter (✓)	
	Translates low-level language to machine code	✓			
	Translates high-level language to machine code		(✓)	✓	
	Produces error messages	(✓)	✓	✓	
	Translates high-level language one line at a time			✓	
	Produces an executable file	(✓)	✓		
1 mark per each correct row:					
NOTE: tick shown in brackets (✓) is optional					

0478/12 – JUNE 2020

Question	Answer	Marks
2(a)	Any four from: – To translate the high-level language into low-level language – Interpreter used whilst writing the program – Interpreter used to debug code line by line – Compiler used when program completed – Compiler used to create separate executable file (so compiler no longer needed) – If it runs first time in a compiler there are no syntax errors	4
2(b)	Any three from: – Easier to understand // Don't know assembly code – Easier to debug – Easier to maintain – Portable – Knowledge of manipulating memory locations/registers not required – Can use an IDE – Greater range of languages	3

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Question	Answer	Marks
8(c)(i)	Any two from: – Paper jam – Out of paper – Out of toner/ink – Buffer full – Awaiting input – Print complete – Printer ready Award any other valid example	2
8(c)(ii)	Any one from: – Operating system – Interrupt handler – Interrupt service routine	1

Question	Answer	Marks
7(a)	Any two from: – Makes use of words // close to human language – Machine independent // portable – Problem / logic focussed – Needs to be translated/interpreter/compiled (to low-level for processing by computer) // needs converting to machine code	2
7(b)	Four from Max 2 for only giving compiler/interpreter features – Both translate high level / source code to machine code – Both generate error diagnostics/messages // identify errors – Interpreter translates one line at a time // checks one line and then runs it – Compiler translates whole code in one go // checks all code and then runs it – Interpreter stops when meets an error – ...and then allows you to continue running from where you stopped // correct errors in real-time – Compiler provides list of all errors – Interpreter does not produce an executable file – Compiler produces an executable file	4
7(c)	Any five from: – Free software is distributed with the source code – ...whereas shareware is not distributed with the source code – Free software allows modification of the application – ...whereas shareware cannot be modified – Shareware normally allows a trial period for the end user // shareware has limited features to start with – Free software is often available free of charge – ...whereas shareware normally has a charge after trial period // shareware has charge to access all features // shareware makes you sign-up/register after trial period // shareware makes you sign-up/register to access all features	5

NOVEMBER 2021 (1)

Question	Answer	Marks
5(a)	Any two from: – Encryption – Biometric device – Firewall – Anti-spyware – Two-factor authentication // two-step verification	2
5(b)	Any two from: – Interrupt / error-handling – Peripheral management – Providing user interface – Platform for running applications // installing / removing software – Manages security // access rights/levels // user account management – Managing time slicing // multitasking	2

Question	Answer	Marks
8(a)	– High-level	1
8(b)(i)	One mark for the correct translator, two marks for the benefit(s). – Interpreter – Easier to debug – ... as errors are immediately reported when detected – Compiler – All errors are reported in a single report – ... meaning they can all be fixed at the same time – No need to recompile code every time a test is run	3
8(b)(ii)	One mark for the correct translator, two marks for the benefits. – Compiler – Creates an executable file – ... so, translator is no longer needed to run it – Source code cannot be stolen // can be provided without the source code	3

NOVEMBER 2021 (2)

Question	Answer	Marks
6(a)	Any one from: – They both translate high-level language into machine code / low-level language – They both check for errors – They both report errors	1
6(b)	Four from (Max 2 per translator): – An interpreter translates and executes the code line by line – ... whereas a compiler translates and executes the whole code all in one go – An interpreter stops translating and reports an error as it finds one – ... whereas a compiler produces an error report at the end of translation – An interpreter does not produce an executable file – ... but a compiler does produce an executable file – An interpreter will execute the code until it finds an error – ... whereas a compiler will not execute any code if there are errors present – An interpreter allows correction of errors in real-time – ... whereas a compiler needs to retranslate the code each time after errors are found and corrected	4
6(c)	– Assembler	1

Question	Answer	Marks
9(a)	Any three from: e.g. – A suitable description of any error that might occur – A peripheral is connected/disconnected – A key on a keyboard is pressed – A mouse button click – A phone/video call is received – A buffer requires more data – A printer has a paper jam – A printer runs out of paper – A printer runs out of ink – When switching from one application to another NOTE: If three suitable different errors are described, this can be awarded three marks.	3
9(b)	Any one from: – The computer would only start a new task when it had finished processing the current task // by example – Computer will not be able to multitask – Errors may not be dealt with – Computer would become impossible to use	1

NOVEMBER 2021 (3)

Question	Answer	Marks
4(a)	Any four from: – Printer generates interrupt – Interrupt is given a priority – Interrupt is queued – Interrupt stops CPU from processing current task – CPU will service interrupt // Interrupt handler services interrupt ... – ... generating an output message to state there is a paper jam	4
4(b)	Any two from: – A suitable description of any error that might occur – A peripheral is connected/disconnected – A key on a keyboard is pressed – A mouse button click – A phone/video call is received – A buffer requires more data – A printer runs out of paper – A printer runs out of ink – Opening an application – When switching from one application to another NOTE: If two suitable different errors are described, this can be awarded two marks	2

Question	Answer	Marks
7(a)	Any four from: – Creates an executable file – ... so, would not release source code – ... so, the source code cannot be stolen/edited. – ... so, would not need to be translated every time // so, translator is not required – ... making it machine independent	4

Question	Answer	Marks
7(a)(i)	1 mark for when e.g. • Development // when writing the program // when debugging 1 mark for explanation to max 2 from: e.g. • ... easier to debug • ...stops when an error is detected • ...reports one error at a time • ...can correct errors in run-time // correct the line and then continue running from that point • ...can test one section without the rest of the code being completed	3
7(a)(ii)	1 mark for when e.g. • After completion // For distribution // For final/repeated testing 1 mark each to max 2 from: e.g. After completion • It creates an executable file • ...than can be distributed without source code • ...so that other people cannot edit/view the code • ...so end users do not need translator software // so end users do not need to compile/interpret each time • ...so it is machine/platform independent (usually) In final testing • It creates an executable file • ...do not need to retranslate for each test sequence • ...can test repeatedly with different data faster	3

Question	Answer	Marks
6(a)(i)	• They both report/check for errors	1
6(a)(ii)	Four from (MAX 2 per translator): • An interpreter translates the code line by line (and executes each line immediately) • ... whereas a compiler translates the whole code at the same time (before executing it) • A compiler produces an executable file • ... but an interpreter does not • An interpreter is required to run the code each time if used • ... whereas a compiler is not • An interpreter stops and reports an error as it is encountered • ... whereas a compiler creates a report of all errors at the end of translation • An interpreter will run code up to the point it finds an error • ... whereas a compiler will not run the code at all if an error is found	4

JUNE 2022(2)

Question	Answer	Marks
4(a)	Any three from: • It uses English-like statements • It needs to be converted to machine code (to be processed by a computer) • ... using a translator • It is portable • One line of code can perform multiple commands	3

JUNE 2022(3)

Question	Answer	Marks
8	One mark per each correct term in the correct place • high-level language • line by line • all at once • executable file • is not required • debugging	6

Question	Answer	Marks
2(a)	No mark for choice. Any four from matching choice. High-level • Easier for programmer to read/write/understand/edit • ... therefore, the programmer is less likely to make mistakes // can write in shorter timeframe • Easier to debug // Easier to find/correct errors • ...so, the programmer can find and correct errors in less time • Game will be machine independent // Game will be portable (between hardware) • ...the game can be used on any computer without a need for understanding of the hardware / compilation for that hardware • Programmer can focus on the problem instead of the manipulation of memory/hardware Low-level • More memory/RAM efficient • ... 3D graphics will have high memory consumption anyway • Allows direct manipulation of memory • ... allows for more efficient control/response time • Allows for use of specialised hardware	4
2(b)	Two from for each compiler and interpreter. Compiler • Checks all code before executing any code • Produces error report with all errors found for the whole code (before translating/running any of the code) • Produces executable file Interpreter • Checks/translate one line of code and then executes it before moving on to the next line • Stops when an error is found • ... when corrected the program can be run from the same position // allows error correction in real time	4

JUNE 2023

Question	Answer	Marks
4	One mark for each correct term in the correct place: • System • Application • Operating • Hardware	4

Question	Answer	Marks
7(a)	Any two from: • Close to the language processed by computers • May use mnemonics • An example is assembly language/machine code	2
7(b)	Any two from: • Can directly manipulate the hardware • No requirement for the program to be portable • Program will be more memory efficient • No requirement for a compiler/interpreter • Quicker to execute • Can use specialised hardware	2

JUNE 2023 (2)

Question	Answer	Marks
5(a)	• C	1
5(b)(i)	Any three from: • It translates the (high-level language) to low-level language/object code/machine code • It translates all the code before it is executed • It creates an executable file	3
5(b)(ii)	Any two from: • It creates an error report after trying to compile • ... displaying all errors in the code • ... that require correction before execution can take place	2

Question	Answer	Marks
5(c)	Any three from: e.g. • Code editors • Run-time environment • Built-in interpreter • Error diagnostics • Auto-completion • Auto-correction • Prettyprint	3

Question	Answer	Marks
10(a)	Two from: • System software provides services that the computer requires • ... whereas application software provides services that the user requires One from (system software): • Utility software // by example e.g. defragmentation software, antivirus, firewall • Operating system One from (application software): • Any suitable example of an application e.g. word processor, web browser, video-editing software	4
10(b)	• Secondary storage // HDD // SSD	1

JUNE 2023 (3)

Question	Answer	Marks
4(a)	B	1
4(b)(i)	Machine code // low-level language // object code	1
4(b)(ii)	Interpreter	1
4(b)(iii)	Compiler	1
4(b)(iv)	Compiler	1

Question	Answer	Marks
11(a)	Operating system	1
11(b)	Any one from: - Create a file - Copy a file - Open a file - Close a file - Move a file - Delete a file - Rename a file - Save a file - Sort files	1
11(c)	Any two from: e.g. - Keeping track of the status of each memory location - Managing the movement of data to and from RAM - Checks that processes have enough memory located to them - Makes sure that two processes don't try to access the same memory location - Manage the transfer of pages between virtual memory and RAM - Allows multitasking	2
11(d)	Interrupt	1

Question	Answer	Marks										
3	One mark for each correct term: <table><tr><th>Term</th><th>Description</th></tr><tr><td>hardware</td><td>A collective term for the physical components of the computer system.</td></tr><tr><td>application software</td><td>A type of software that provides services that the user requires and allows the user to perform tasks on the computer.</td></tr><tr><td>operating system</td><td>A type of software that manages the main functions of the computer, including managing files and managing memory.</td></tr><tr><td>firmware</td><td>A type of software that is stored in the read only memory (ROM). It includes the basic input output system (BIOS) and the bootloader.</td></tr></table>	Term	Description	hardware	A collective term for the physical components of the computer system.	application software	A type of software that provides services that the user requires and allows the user to perform tasks on the computer.	operating system	A type of software that manages the main functions of the computer, including managing files and managing memory.	firmware	A type of software that is stored in the read only memory (ROM). It includes the basic input output system (BIOS) and the bootloader.	4
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Question	Answer	Marks
9(a)	Any one from: – Operating system // Interrupt handler	1
9(b)	Any five from: e.g. – Key press generates the interrupt – Interrupt given a priority – Interrupt is sent to CPU – Interrupt is placed in a queue – CPU stops current task to check the queue/service the interrupt ... – ... using an interrupt service routine – If key press is highest priority the interrupt is processed	5
9(c)(i)	Any two suitable hardware example e.g.: – Moving the mouse – Clicking a mouse button – Plugging in a device – Paper jam in printer – Printer out of paper	2
9(c)(ii)	Any two suitable software examples e.g.: – Division by zero – Two processes accessing the same memory location – Null value	2

NOVEMBER 2023 (3)

1(d)	Any three from: – It performs the basic functions of a computer – It manages the hardware – It provides a platform to run software – It provides a user interface – It performs tasks such as (any example of function of an operating system)	3
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